Proposal for ReportNow: A web reporting service for issues in Singapore

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Submitted to—
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Table of Contents

Executive Summary

Singapore's Smart Nation initiative emphasizes citizen-centric services, yet residents still face difficulty identifying the correct agencies to report issues such as potholes or damaged infrastructure. To address this, our team proposed the development of ReportNow, a web and mobile-based platform enabling citizens to submit and track reports while automatically routing them to the relevant government agencies.

The platform would incorporate GPS-based location reporting, guardrails in user registration to ensure system robustness and security, and a user-friendly interface. It could further leverage AI to categorize issues by importance or detect duplicates, while features such as crowdsourced ranking through user upvotes, manual reclassification, and administrative recategorization would enhance both accuracy and responsiveness.

The system will adopt a Model–View–Controller (MVC) layered architecture, employing React/Next.js, Node.js, Prisma, and PostgreSQL, while integrating AI through the OpenAI API. These improvements will produce a more reliable and user-friendly platform that supports Singapore's Smart Nation 2.0 vision of inclusive, transparent, and citizen-driven digital services.

Statement of Problem

Singapore's Smart Nation vision, launched in 2014, aimed to use technology to improve citizens' lives (Ministry of Digital Development and Information [MDDI], 2024, p. 2). While significant progress has been made, the refreshed Smart Nation 2.0 acknowledges gaps that remain as well as new opportunities and challenges that have emerged (MDDI, 2024, p. 3). This creates an opening to harness digital technology for more inclusive, responsive, and community-focused solutions under the Smart Nation 2.0 framework.

Despite strong advances in its digital sector, the Singapore government recognises the need to continuously examine the societal impact of technology (MDDI, 2024, p. 25). Existing platforms such as the OneService and Kaki chatbot enable municipal issue reporting, with AI routing feedback to relevant agencies (MDDI, 2024, p. 15). Hackathons such as Build for Good also promote citizen co-creation of digital solutions (MDDI, 2024, p. 79). These efforts highlight the government's commitment to community-driven innovation.

However, a key challenge remains in the absence of a universal, intuitive platform that can manage a wider spectrum of public concerns, such as accidents or infrastructure faults. Existing solutions like OneService remain primarily municipal in scope, leaving room for more expansive and integrated systems. This gap underscores the opportunity for new approaches that bridge current limitations, foster inclusivity in community participation, and enhance transparency and responsiveness in addressing community issues.

Objectives

This document proposes ReportNow, a web and mobile-based platform for reporting issues in Singapore. The objectives of this project are as follows:

(1) Build a robust reporting platform that allows citizens to submit, track, and manage a broad range of public concerns.

(2) Deliver an intuitive, user-friendly interface to ensure accessibility and encourage widespread adoption.

(3) Implement intelligent issue management capabilities, including AI-assisted categorisation, duplicate detection, and crowdsourced feedback, to enhance accuracy and responsiveness.

The system will display the most recent reports on first load. User registration will include strong validation measures to ensure reliable participation, rejecting invalid entries such as incorrect email formats or incomplete fields.

The reporting workflow will be streamlined so that new users can submit an issue by inputting title, description, location and optional image upload without unnecessary complexity. Navigation will be designed for efficiency, allowing users to access detailed report information within three taps or clicks from the home screen.

The system will incorporate an AI model trained to categorise issues into predefined types (e.g., potholes, accidents, faulty infrastructure) with a target classification accuracy of at least 80% on labelled data. A duplicate detection mechanism will flag similar or identical issues within seconds of submission, reducing redundancy. Users will also be able to upvote reports, creating a real-time crowdsourced ranking that highlights issues of greatest collective concern.

Technical Approach

This section outlines how the project objectives will be achieved through a structured process of design, development, and deployment. The proposed approach focuses on building a robust reporting platform, with emphasis on core functionality, user-friendly design, and intelligent issue management capabilities that align with Singapore's Smart Nation 2.0 goals.

Customer Needs

Citizens require a single, user-friendly platform for reporting a wide range of public issues without having to determine which government department is responsible. Government departments require a system that categorizes, routes, and tracks issues efficiently while providing them with administrative controls. The proposed system also needs to meet modern expectations for usability, accuracy, transparency, and responsiveness, aligning with Singapore's Smart Nation 2.0 vision of inclusive, citizen-centric digital services.

Target Specifications

To ensure that the proposed system meets the needs of both citizens and government departments, the design will adhere to measurable specifications.

(1) The reporting platform will be optimized to display the newest issues at the top, with older reports appearing below, ensuring a clear and intuitive chronological order for users.

(2) Location accuracy will be refined to within approximately ten meters on supported devices when connected to campus WiFi, ensuring that reported issues can be precisely mapped.

(3) User registration will incorporate validation guardrails designed to reject at least 95% of invalid input such as incomplete forms or incorrectly formatted email addresses, thereby strengthening data integrity.

(4) Newly submitted reports are expected to appear on the map within three minutes of submission, allowing near real-time feedback to users.

(5) The AI classification model must achieve a minimum accuracy of 80% on a labeled test dataset, and duplicate detection should be able to flag potentially similar or identical reports within five seconds of their creation.

Technology Consideration

The proposed system will adopt a modern web-based technology stack to ensure scalability, cross-platform accessibility, and ease of maintenance, while enabling faster updates and integration with emerging technologies.

The presentation layer will be developed using Next.js and React to deliver a responsive and modular interface, with TailwindCSS, HTML, and CSS providing a consistent design framework.

The application layer will be powered by Node.js and serverless API routes deployed on Vercel, while NextAuth.js will manage authentication, including integration with Singpass login.

Within the business layer, JavaScript will be used for implementing logic, supplemented by the OpenAI API to enable AI-based categorization and duplicate detection of reported issues.

Automated testing will be supported by Jest and React Testing Library to ensure functionality and reliability.

For the persistence layer, Prisma ORM will be employed to streamline schema definitions and database queries, with PostgreSQL serving as the relational database for secure storage of issue reports, user accounts, and system logs.

Dependency management and security scanning will be handled by Dependabot and Scantist to ensure ongoing maintainability and resilience against vulnerabilities.

System Architecture/Platform

The proposed system will adopt a Model–View–Controller (MVC) architecture to ensure a clear separation of concerns, supporting maintainability, scalability, and efficient development.

At the view layer, users interact through dedicated interfaces, including the CitizenUI, GovUI, AdminUI, SingpassUI, and a central MainUI that serves as the entry point. These views are tailored to role-specific functions, such as citizen issue reporting, government account management, or administrative oversight.

The controller layer manages the flow of data and system logic. Controllers such as the Login Controller, Gov Controller, Citizen Controller, Admin Controller, and Issue Controller act as intermediaries between the user interfaces and the underlying models. Specialized controllers, such as the AI Categoriser Controller and AI Chatbot Controller, extend system functionality by providing intelligent issue classification and conversational support.

The model layer represents the core entities of the system, including Citizens, Government Departments, Admins, Issues, Reviews, and Government Accounts. These models encapsulate the data and rules associated with each domain entity. For example, the Issue model links directly to both citizen reports and government updates, while the Review model enables feedback and evaluation once an issue has been resolved.

This MVC-style architecture promotes modularity and scalability. Each layer operates independently, allowing modifications to be made to one component (e.g., adding a new issue classification algorithm in the Issue Controller) without disrupting the rest of the system. The architecture also supports role-based access control, where users access only the functionality relevant to their role. This design ensures maintainability, facilitates debugging, and provides a clear foundation for iterative enhancements such as improved AI classification, duplicate detection, and UI refinement.

Project Management

The breakdown of the development phases for the project are as follows:

1. Planning: The project, with regards to the various aspects of the problem at hand, is to be discussed amongst team members to formulate a concrete plan on its development stages with estimates on the relevant deadlines. This includes the division of work and determining the target audience.

2. Concept Development: The project's foundation is to be set with its base functionality, platform, modularization and presentation with regards to the purpose and relevant target audiences for the project.

3. System-Level Design: The design for the project at its system level, with functional codes that simulate the desired functions for each module of the program is to be built and tested accordingly.

4. Detailed Design: The fragments of the project built in the preceding stage are to be assembled together, with frontend integration. Minor fixes and changes will be accounted for in this stage, and modularity checks will be conducted for the convenience of future modifications and refinements.

5. Testing and Refinement: The functionality of the program will be tested under various use cases, with extreme and abnormal inputs to pre eliminate errors and system hangs caused by invalid inputs. The project will be refined based on the tests conducted.

6. Production: The project will be pushed to production after sufficient testing and refinements. The project will be continued to be monitored for maintenance and upgrades as required.

The division of responsibilities and duties among team members are as follows:

Project Manager: Madelyn

QA Manager: Yong Huat

QA Engineer: Kean Yee

Lead Developer: Wei Ming

Front-end Developer: Dylan

Back-end Developer: Seo Hyeon

Release Manager/Engineer: Jun Howe

The division of responsibilities and duties are flexible; each member is entrusted with their own sector of the project, but will not be limited to the domain and collaborate mutually for the successful completion and delivery of the project. The proposed timeline is based upon the prototype deliverable date of October 9, 2025. Tasks are based off of the required technical aspects of the project as reflected within the backlog.

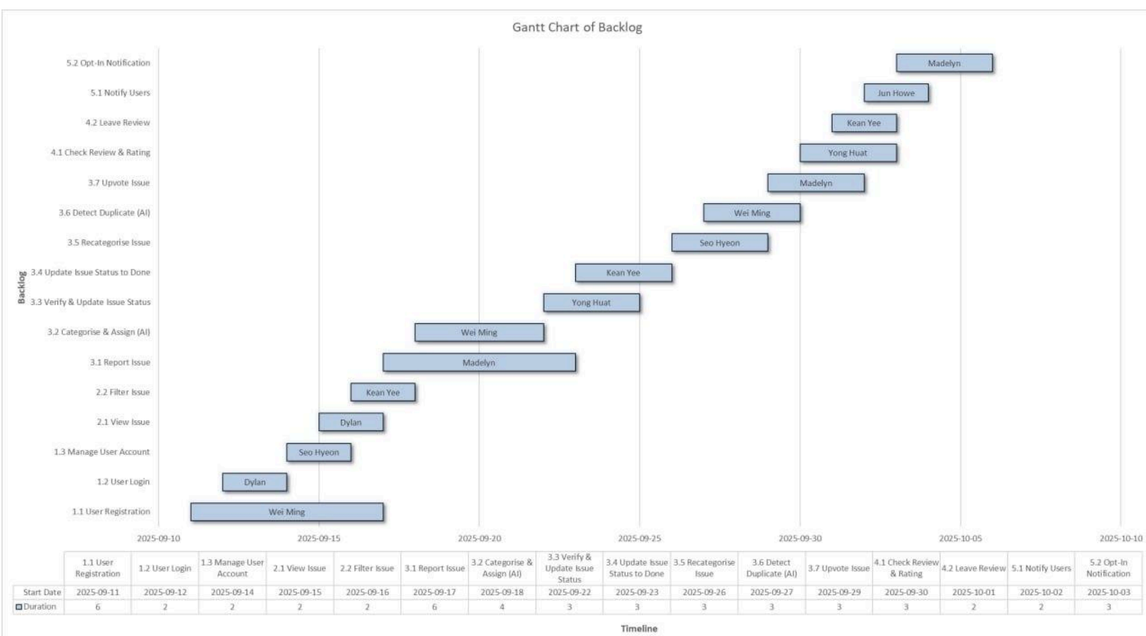


Figure 1: Gantt chart of initial project backlog

Deliverables

As per the Gantt chart, our team will attempt to adhere to the schedule as planned as closely as possible, preparing and completing all the required deliverables for the lab sessions as we progress our development of the project.

The deliverables for this project are defined by the weekly lab manuals issued under SC3040 Advanced Software Engineering. Each lab session specifies required outputs, which typically include documentation, design artifacts, source code, and progress reports. These deliverables will be prepared collaboratively by the team and submitted through the project's GitHub repository, which serves as the central platform for code integration, documentation, and version control.

Final grading and evaluation of the deliverables will be carried out by the teaching assistant, Mr. Kalkauskas Antanas, based on the quality, completeness, and timeliness of each submission. By aligning deliverables with the structured requirements of the lab manuals, the team ensures that work is well scoped, trackable, and aligned with the course objectives.

Budget

The budget required for this project is negligible, and only includes development-related token costs through API calls to a large language model. Costs to be incurred are downwards of a few cents.

Communication and Coordination with Sponsor

As this is an academic project, the primary sponsor role is represented by the course instructors and the assigned teaching assistant, Mr. Kalkauskas Antanas. Communication with the sponsor will be maintained on a bi-weekly basis through lab sessions, where progress is reviewed, questions are clarified, and feedback is provided. Additional updates or clarifications will be shared through the course's official communication channels, such as email or NTUlearn, as necessary.

All team members are responsible for keeping track of their assigned tasks and reporting progress during scheduled meetings. The team leader will consolidate updates and ensure that important questions or blockers are raised with the sponsor during lab sessions. Information transmitted in weekly communications is primarily for progress reporting and clarification, with feedback from the sponsor taken as action items to be addressed in the following iteration of work.

Team Qualifications

The project team consists of Year 3 Computer Science students from the College of Computing and Data Science at Nanyang Technological University, working together under SC3040 Advanced Software Engineering (TEL5, Team 2). Collectively, the team has a strong foundation in core areas of computer science, including databases, software engineering, artificial intelligence, and human-computer interaction. Each member brings complementary skills in areas such as frontend and backend development, user interface design, system integration, and project coordination, enabling the team to address both the technical and management aspects of this project. Brief descriptions of each member's qualifications are provided below, with full resumes included in Appendix A.

Dylan Quek Zhi En

Dylan Quek Zhi En is a Year 3 Computer Engineering student in the College of Computing and Data Science at Nanyang Technological University, currently enrolled in SC3040 Advanced Software Engineering as part of Team 2 (TEL5). He has gained relevant experience through coursework in Multidisciplinary Project and Software Engineering, as well as through project work involving data science and web development. For this project, Dylan contributes strengths in web development and UI/UX design. These skills, combined with academic training, enable him to support the design, development, and delivery of the ReportNow. A complete resume is provided in Appendix A.

Ang Wei Ming

Ang Wei Ming is a Year 3 computer science student, and he has experience been a full-stack developer and a product manager. He has built a few mobile applications and websites used by a few thousand users before. He has experience in using Next.js, React last time, Node.js, Payload, Flutter, Sequelize, MongoDB, Firebase, AWS, and Vercel, just to name a few. As a developer, he finds it important to prioritise user experience, security, performance and scalability when developing. Beyond coding, he mentors other aspiring developers in websites he develops for his school and drives community engagement through volunteering at community events. For this project, Wei Ming contributes strengths in web development and UI/UX design. These skills, combined with academic training, enable him to support the design, development, and delivery of the ReportNow. A complete resume is provided in Appendix A.

Chi Seo Hyeon

Chi Seo Hyeon is a Year 3 Computer Science student in the College of Computing and Data Science at Nanyang Technological University, currently enrolled in SC3040 Advanced Software Engineering as part of Team 2 (TEL5). He has gained relevant experience through coursework in Object Oriented Programming, Multidisciplinary Project, Software Engineering. For this project, he contributes as the backend developer, responsible for designing, building, and maintaining the server-side logic, databases, and APIs, ensuring high performance, scalability, and seamless integration with the front-end applications. These skills, combined with academic training, enable him to support the design, development, and delivery of the ReportNow. A complete resume is provided in Appendix A.

Choo Kean Yee

Choo Kean Yee is a Year 3 Computer Science student in the College of Computing and Data Science at Nanyang Technological University, currently enrolled in SC3040 Advanced Software Engineering as part of Team 2 (TEL5). He has gained relevant experience through coursework in Databases, Artificial Intelligence, as well as through project work involving web development and automations. For this project, Kean Yee contributes strengths in backend integration and QA. These skills, combined with academic training, enable him to support the design, development, and delivery of ReportNow. A complete resume is provided in Appendix A.

Madelyn Cruz Tan

Madelyn Cruz Tan is a Year 3 Computer Science undergraduate in the College of Computing and Data Science at Nanyang Technological University, currently enrolled in SC3040 Advanced Software Engineering as part of Team 2 (TEL5). On top of her relevant courses taken throughout the duration of her candidature which includes, Software Engineering, Introduction to Databases, Artificial Intelligence, etc., She has had experience as a full stack developer as an independent contractor for [REDACTED], working on the company's CRM. She is also in the process of attaining a Deep-Learning Specification externally, and liaising [REDACTED]'s website development with an outsourced company. For this project, Madelyn's contribution as project manager and member of the team will be to keep the team on track while contributing to the technical and administrative aspects of the project. A redacted version of her resume is provided in Appendix A.

Lim Jun Howe

Lim Jun Howe is a Year 3 Computer Science student in the College of Computing and Data Science at Nanyang Technological University, currently enrolled in SC3040 Advanced Software Engineering as part of Team 2 (TEL5). He has gained relevant experience through coursework in Software Engineering and Multidisciplinary Project. For this project, he contributes as the Release Engineer, ensuring a smooth deployment process for our web application, as well as building and integrating changes for delivery.

These skills, combined with academic training, enable him to support the design, development and delivery of ReportNow. A complete resume is provided in Appendix A.

Tan Yong Huat

Tan Yong Huat is a Year 3 Computer Science student in the College of Computing and Data Science at Nanyang Technological University, currently enrolled in SC3040 Advanced Software Engineering as part of Team 2 (TEL5). He has gained relevant experience through coursework in Object Oriented Programming, Multidisciplinary Project, Software Engineering. For this project, he contributes as the Quality Assurance Manager, responsible for ensuring the system meets defined requirements through systematic testing, validation, and continuous feedback to the development team. These skills, combined with academic training, enable him to support the design, development, and delivery of the ReportNow. A complete resume is provided in Appendix A.

References

Ministry of Digital Development and Information (MDDI). (2024, October). *Smart Nation 2.0: A thriving digital future for all*. Retrieved September 8, 2025, from https://file.go.gov.sg/smartnation2-report.pdf

Appendix A: Résumés of Team Members

The following pages present one-page résumés of the team members for this project.

Dylan Quek Zhi En

Dylan Quek | Mobile No.: 81301116 | Email: quekdylan7@gmail.com

EDUCATION

Nanyang Technological University, Singapore Aug 2023 - May 2027
Bachelor of Engineering in Computer Engineering

Ngee Ann Polytechnic, Singapore Apr 2018 - Apr 2021
Diploma in Information Technology

- Edusave Certificate of Academic Achievement 2018, 2019

EXPERIENCE

PSA Singapore Mar 2020 - Jul 2020
Data Scientist

- Wrote python scripts to prepare quay crane sensor data for machine learning algorithms, for planning of upcoming Tuas megaport
- Manipulated SQL queries to help migrate reports to a new database. Gained experience in SQL syntax and Microsoft Access

SKILLS

- **Programming Languages:** Python, Java, C++, C, C#, SQL, Assembly
- **Web & Mobile Development:** React, React Native, Tailwind CSS, HTML, CSS, JS
- **Machine Learning / Data Science:** Python, Pandas, NumPy, Scikit-learn
- **Tools & Hardware:** Linux CLI, STM32 MCU, Raspberry Pi

PROJECTS

ParkIT Jan 2025 – May 2025
Frontend Developer

- Built a web-based app that connects with several APIs to provide live carpark data in Singapore
- Gained experience in React framework, Java Spring Framework, VPS hosting

Galaxias Staking Mar 2022 – Mar 2024
Founder

- Developed and maintained 3 validator nodes for various Tendermint and Ethereum based block-chains
- Built website landing page showcasing features and information for 3 chains used
- Gained experience on Linux CLI, web development, Grafana dashboards and blockchain

Dengue.live Oct 2019 - Mar 2020
Web developer, Project Lead

- Built a website with map-based dashboard showing predicted future dengue clusters using machine learning
- Gained experience on AGILE framework and HTML, CSS and Javascript

COMMUNITY SERVICE

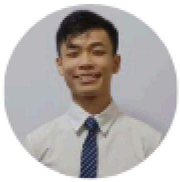
Handicaps Welfare Association Jan 2014 - Dec 2017
Packer

- Performed various tasks, including cleaning and packing at the Handicaps Welfare Association.

INTERESTS

- Web development, UI/UX design, Blockchain, Crypto

Ang Wei Ming



ANG WEI MING

FULL-STACK DEVELOPER
COMPUTER SCIENCE STUDENT

angwm555@gmail.com
+65 86601646
https://www.linkedin.com/in/ang-w-391a8b124

EDUCATION

Nanyang Technological University, Singapore
Bachelor of Computer Science
- Expected Minor in Entrepreneurship

Singapore Polytechnic
Diploma Information Technology
Diploma with Merit
- *DIST Mathematics
- *DIST Fundamentals of Programming
- *DIST Communicating for Project Effectiveness
- *DIST Critical and Analytical Thinking
- *DIST Fundamentals of Programming
- *DIST Java Programming

GCE O Level Certificate
(Catholic High)
- L1R5: 5

EXECUTIVE PROFILE

Determined, Innovative, Goal-Oriented.

CAREER OBJECTIVE

Have at least ONE important takeaway from every job

PROFESSIONAL EXPERIENCE

Full-Stack Developer at Venturbuild
FEB 2025 - CURRENT
- Developing a website that connects researchers at Institutes of Higher Learning with Entrepreneurs

Technical Lead at NTU
DEC 2024 - FEB 2025
- Developed a website for the NTU Product Club that promotes its product management course, events collaborations/hackathons/podcasts
- Website made with NextJS, Supabase, Vercel, Tailwind
- Led a new team of 4 developers to become proficient in web development skills

Technical Lead at DoctorNowSG (Startup)
JUL 2024 - DEC 2024
- Developed a medical web app to boost medical tourism in Singapore with international translations, reaching clients across 3 countries
- Web app made with Flutter and Firebase, available on App Store, Google Play Store & Internet Browser

Technical Lead at Kemp Studios (Startup)
MAY 2024 - JUL 2024
- Developed walking event mobile app for Jack Neo PaPaZao initiative has over 3K weekly users
- Mobile app made with Flutter and Firebase, available on App Store & Google Play Store

Full Stack Developer at Associates Consulting
FEB 2021 - AUG 2022
- Digitalised ISO 9001:2015, created a content management system with a team of developers
- Created customisable multi-level permissions control, e.g. different access scopes for admins
- Website made with React, NodeJS, AWS, SASS

Contractor at Anacle Systems Pte Ltd
NOV 2019 - MAR 2020
- Promoted from Part-time job to contractor due to outstanding performance at work
- Streamlined workflow through innovation
- Led a team to complete over 3K+ orders involving delivery, door-to-door sales, telemarketing, admin paperwork, inventory handling, database validation

Chi Seo Hyeon

Chi Seo Hyeon

+65 9065 9858 · shc.leinad@gmail.com
38 Nanyang Crescent, 636866

PROGRAMMING AND DATABASE MANAGEMENT

Motivated and solution-oriented programmer with elaborate experiences in back-end program and database management. Stationed as a military interpreter for the South Korean National Service, led and managed a foreign intelligence collection platoon focused around military intelligence and translation, building experiences in linguistics and programming along with skills revolving around the maintenance of complex database systems.

KEY COMPETENCIES

Back-end programming with proficiency in Python, Java, C and PHP

Database management skills with relevant experiences in SQL, Excel and MS Access

Fluent communication skills and solution-oriented leadership with experiences in military intelligence

PROFESSIONAL EXPERIENCE

KDIC **Nov 2023 - Mar 2025**
Military interpreter
Maintained and refined the military translation program for intelligence collection as the leader of the foreign intelligence platoon in the Republic of Korea Army. Maintained a military database throughout the service, and led multiple English to Korean translation projects.

PROJECTS

Parking Record Management System **January 2023 - August 2023**
Project Manager
Led the team in creating a parking management database system for all public parking spaces in Singapore. Overlooked the back-end to front-end integration and specialized in program modularization and production during the development phases as the project manager.

AI translator project **August 2022 - December 2023**
Project Manager
Led the team in developing an AI based translator web application for a more accurate translation of internet slangs and abbreviations from English to Korean and back.

EDUCATION & CERTIFICATIONS

Bachelor of Computer Science
Nanyang Technological University
Singapore

EXTRACURRICULAR ACTIVITIES

President, Debate Club 2019 - 2021
Gandhi Memorial International School, Jakarta

Vice President, Programming Club 2018 - 2020
Gandhi Memorial International School, Jakarta

Choo Kean Yee

Choo Kean Yee | Mobile No.: +65 94781748 | Email: keanyeechoo@gmail.com

EDUCATION

Nanyang Technological University (NTU), Singapore Aug 2022 - May 2026
Bachelor of Engineering in Computer Science

EXPERIENCE

Central Provident Fund (CPF) Board - TechOps Intern **July 2024 - current**

- Developed an internal retrieval-augmented generation (RAG) chatbot equipped with domain knowledge to improve efficiency in handling internal queries and onboarding.
- Kickstarted the Confluence project to centralize documents and scattered knowledge, integrating them into a common knowledge base that feeds into the RAG vector database, enhancing the chatbot's effectiveness.
- Designed intuitive landing pages for the Confluence project using HTML/CSS, ensuring ease of access and usability for internal users.

PROJECTS

AI Chatbot Developer - HealthcareGoWhere App **Mar 2024**
Google AI/ML Hackathon for Sustainability and Social Inequality

- Collaborated in a diverse team to develop "Healthcare GoWhere," a web application aimed at addressing healthcare accessibility challenges.
- Spearheaded the integration of an AI-powered chatbot that assists users by locating nearby clinics and visualizing their positions on a map, enhancing user experience and accessibility.
- Utilized technologies such as Large Language Models (OpenAI API) and Google Maps API to design and implement the chatbot's natural language processing and geolocation mapping features

Software Engineer (Project Team Leader) - TransiTrack App **Jan - Apr 2024**
NTU, Software Engineering Project

- Developed a user-centric transportation app utilizing React for dynamic frontend interactions, enhancing user experience by providing real-time bus arrivals, route recommendations, and crowd level indicators.
- Integrated Google Maps API and LTA DataMall API to fetch accurate geolocation data and real-time bus stop information, ensuring up-to-date transit details and enhancing route accuracy.
- Engineered robust backend solutions with Express, handling complex queries and integrations, which significantly increased the app's response time and reliability.
- Implemented Firebase database management to store and retrieve user data effectively, ensuring high availability and consistency of transit information.

March Madness Analysis - Team Leader **Jan - Mar 2023**
NTU, Data Science and Artificial Intelligence Project

- Utilised Jupyter Notebook to analyse a database on March Madness and improve prediction accuracy
- Automate data collection and aggregation processes for data cleaning, data analysis and data visualisation
- Implemented various machine learning models to determine how much certain statistics impacted winning

Basketball Court Booking Automation Bot - Developer **Sept 2024**
NTU Hall 3 Basketball Captain

- Developed an automation bot to streamline the weekly booking of NTU basketball courts for my hall using Cursor.ai and Selenium.
- Implemented web scraping and browser automation to manage time-sensitive bookings efficiently, ensuring court availability for team practices.
- Reduced manual effort and increased booking success rate, allowing consistent weekly training schedules.

RELEVANT COURSEWORK

- Data Structures and Algorithms
- Object Oriented Design and Programming
- Software Engineering
- Databases
- Data Science and Artificial Intelligence

TECHNICAL SKILLS

- Programming Languages: Python, C, Java, HTML/CSS, SQL, JavaScript, Vue, React
- Tools: Jupyter Notebook, IntelliJ, VSCode, GitHub, Figma, Firebase, Selenium, Cursor.ai, Replit

Madelyn Cruz Tan

NAME: Madelyn Cruz Tan | **Mobile No.: +65 91747007 | Email: madelyntan0223@gmail.com**

EDUCATION

Nanyang Technological University — Aug 2023 - Current

MINDEF/SAF/ARMY Logistics Training Institute — Nov 2021 - Nov 2021
Diploma

Temasek Junior College — Jan 2020 - Dec 2020
GCE "A" Levels

SKILLS

- Programming: Python, Java, C++, JavaScript, SQL
- AI & Machine Learning: TensorFlow, PyTorch, Scikit-learn
- Web Development: Django, React.js, Pygame, Next.js, Tailwind.css, etc.
- Data Science: Pandas, NumPy, Data Preprocessing
- Soft Skills: Leadership, Problem-Solving, Team Collaboration

RELEVANT EXPERIENCE

Independent Contractor at [REDACTED] — July 2025 -- Current
- Development of company CRM
- Liaison with outsourced web-development team for company website
- IT Lead for company backend systems

TechFest NTU Hackathon 2025 — Jan 2025
- Developed a telegram bot to detect fake news
- The bot also learns known misinformation, and stores it for quick reference
- Telegram, OpenAI, Gemini, Google APIs, sqlalchemy

Neural Networks and Deep Learning — Jan 2024 – Dec 2024
- Completing Andrew Ng's Deep Learning Specialization on Coursera (**4/5 courses**)
- Built and optimized Neural Networks from scratch, implementing hyperparameter tuning techniques
- Applied regularization techniques to improve model performance

WiDS Datathon Challenge #2
Team of Three
- Preprocessed and analysed 10,000+ data points using Python libraries
- Developed predictive models using Scikit-Learn
- Applied data handling techniques to clean and process large datasets

OTHER EXPERIENCE

Hall 8 - NTU
Chess Team Manager — Aug 2024 – Current
- Trained 8 players and achieved Top 16 in the Inter-Hall competition.

Freshmen Orientation Committee Programmer
- Planned 3-day orientation for incoming hall freshmen

Epilogue Props Subcommittee
- Hall 8's theatrical productions subcommittee in charge of props of scenes

Carrom Top 16
- Achieved Top 16 in Carrom Inter-Hall competition

Cultural Activities Club - NTU — Jan 2023 - Dec 2024
Logistics Director
- Managed venue bookings and logistical support for 30+ arts, music, and performance clubs
- Led the planning and execution of NTU's Annual Festival Event: CAC Welcome Tea, coordinating performances, vendors, and 500+ attendees

Lim Jun Howe

Lim Jun Howe

+65 8688 4877 · limjunhowe@gmail.com

TECHNICAL SKILLS

- **Programming Languages:** Python, Javascript, C
- **Cloud:** AWS, Azure, Vercel
- **Development Tools:** Git, Docker
- **Front-end Frameworks:** NextJS, React, Bootstrap
- **Databases:** MySQL, MongoDB, DynamoDB

UNIVERSITY PROJECTS

Car Kaki

- Developed a smart car park web application with real-time parking availability monitoring, interactive map interface with Google maps integration, turn-by-turn navigation, dynamic parking rate information and secure user authentication.
- Made Car Kaki cross-platform, compatible with Windows, Mac and Linux for desktops, iOS and Android for mobile devices. Can be used with Apple Carplay and Android Auto while in vehicles as well.
- Deployed Car Kaki on Vercel's serverless platform (Link: car-kaki.vercel.app)

Hospital Management System

- Developed a comprehensive Java-based command line interface application for managing hospital operations that features a custom CSV-based persistence layer and automated data initialisation from Excel files.
- Implemented a multi-role user system that includes a patient portal, doctor's interface, nurse's interface, pharmacist dashboard and admin console.
- Enhanced system architecture by implementing generic repositories, a robust error handling system and a data validation framework.

Lego Set Price Prediction

- Performed exploratory data analysis on lego price datasets over the years after merging and cleaning them, extracting and individually analysing key features such as rating, availability and recent sale prices.
- Created correlation analysis pipeline to identify the most significant relationships between our identified key features.
- Implemented a model performance evaluation system to monitor the accuracy of predictions over time.

WORK EXPERIENCE

System Network Administrator, Singapore Armed Forces — **Aug 2021 - Jan 2023**

- Specialised in computer information and communication systems to set up, manage and maintain computer networks that were critical in the preparation and enactment of successful military operations.
- Handled administrative tasks which included in-processing and out-processing of company personnel, as well as tracking documents that were classified confidential and above.

Output Quality Checker, United Overseas Bank — **Feb 2023 - Jun 2023**

- Ensured that various prepared documents were of good quality and packed them to be delivered to a storage facility.
- Indented supplies and logistics quarterly to facilitate smooth office operations.
- Handled recalling of documents from storage facility when necessary.

EDUCATION & CERTIFICATIONS

Bachelor of Computing, Computer Science
Nanyang Technological University

- Expected Graduation: 2027

Tan Yong Huat

Tan Yong Huat | **Mobile No.: 83995697 | Email: yonghuat12@gmail.com**

PROFESSIONAL SUMMARY

Computer Science student with hands-on experience in software development, robotics, and product design. Proficient in frontend development with strong problem-solving abilities, effective communication skills, and the ability to work well in multidisciplinary teams.

EDUCATION

Nanyang Technological University, Singapore — Aug 2023 - May 2027
Bachelor of Computing in Computer Science

Singapore Polytechnic, Singapore — Apr 2017 - Mar 2020
Diploma in Mechanical Engineering (with Merit)

ACADEMIC PROJECTS

Nanyang Technological University, Robotic Car Multi-Disciplinary Design Project — Aug 2025 - Present
Student

- Currently developing grid-based pathfinding algorithms such as A* for the robotic car, requiring comprehensive understanding of entire system architecture to integrate computer vision and sensor fusion components.
- Collaborate with multidisciplinary team to integrate hardware systems, Android mobile interface, and Raspberry Pi controllers for robotic operation.

Nanyang Technological University, Carpark & Express-Way Application — Jan 2025 – Apr 2025
Student

- Developed favoriting functionality and user interface using Android Studio, managing complex integration between carpark and expressway data systems.
- Collaborated with team to implement SQL database for persistent data storage and applied software engineering methodologies including UML modeling (Use-Case, Sequence, and Class Diagrams).
- Built reusable UI components to enhance user experience and streamline development process.

Nanyang Technological University, Hospital Management System — Aug 2024 - Nov 2024
Student

- Designed and implemented Doctor module for multi-user Hospital Management System serving patients, doctors, administrators, and pharmacists.
- Implemented core Object-Oriented Programming (OOP) concepts (inheritance, polymorphism, encapsulation) and design patterns in Java development.

Singapore Polytechnic, Year 3 Final Year Project, Duct Cleaning Robot — Aug 2019 - Feb 2020
Student

- Designed and prototyped a duct cleaning robot with integrating pneumatics and electrical components.
- Created 3D models and assembly drawings using Autodesk Inventor for manufacturing and component integration.
- Programmed Arduino-based control system for automated cleaning operations and sensor feedback.
- Established foundational design framework and collected critical experimental data for future robot development iterations.

Singapore Polytechnic, Product Design Course, EcoScooter — Feb 2019 - Aug 2019
Student

- Conducted comprehensive market research and SWOT analysis to identify sustainable transportation opportunities
- Designed innovative solar-powered e-scooter concept integrating photovoltaic films for renewable energy harvesting
- Created visually compelling 3D product visualization and marketing materials using Cinema4D and Adobe Illustrator, focusing on aesthetic appeal to enhance potential user engagement

TECHNICAL SKILLS

- Programming Languages: Java, Python, C, C++
- Frontend Development: Figma, Android Studio, UI/UX design principles
- CAD/3D Modelling: Autodesk Inventor, AutoCAD, Inventor CAM, Maya, Cinema 4D
- Digital Media: Adobe Photoshop, Illustrator, Premiere Pro
- Product Development: Research, design, component selection, prototyping, fabrication, testing
- Manufacturing Equipment: CNC machines, mills, lathes, 3D printers